FLP

I Unificare

x, y, z, v, u – var. $\quad a, b, c$ – constante

① $p(a, x, h(g(y))), \; p(z, h(z), h(u))$

Sol	Rez	Op
$\emptyset$	$p(a, x, h(g(y))) = p(z, h(z), h(u))$	Desc
$\emptyset$	$a = z, \; x = h(z), \; h(g(y)) = h(u)$	Desc
$\emptyset$	$a = z, \; x = h(z), \; g(y) = v$	Rez
$z = a$	$x = h(a), \; g(y) = v$	Rez
$x = h(a), \; z = a$	$g(y) = v$	Rez
$v = g(y), \; x = h(a), \; z = a$	$\emptyset$	Stop

$\sigma = \{ v/g(y), \; x/h(a), \; z/a \}$ → CGU

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② $f(h(a), g(x))$ și $f(y, y)$

Sol ← DUPĂ „REZOLVĂ" și modifică Rezolvare

~~∅~~	$f(h(a), g(x)) = f(y, y)$	Op
~~∅~~	$h(a) = y,\ g(x) = y$	DESC
	$g(x) = h(a)$	~~REZ~~
$y = h(a)$		EȘEC

⇒ ~~(2)~~ ~~ecuație~~ unificator

③ $p(a, x, g(x))$ și $p(a, y, y)$

SOL	~~REZ~~	Op
~~∅~~	$p(a, x, g(x)) = p(a, y, y)$	DESC
~~∅~~	$a = a,\ x = y,\ g(x) = y$	SCOATE_
~~∅~~	$x = y,\ g(x) = y$	~~REZOLVĂ~~
$x = y$	~~$g(x) = x$~~ $g(y) = y$	EȘEC.

⇒ ~~(3)~~ unificator

④ $p(x, z, z)$, $p(y, y, b)$

SOL	~~REZ~~	Op
~~∅~~	$p(x, z, z) = p(y, y, b)$	DESC.
~~∅~~	$x = y,\ z = y,\ z = b$	~~REZ~~
$x = y$	$z = y,\ z = b$	REZ
	$y = b$	~~REZ~~
$z = y,\ x = y$		STOP
$y = b,\ z = b,\ x = b$		

~~(2)~~ $\lambda = \{y|b, z|b, x|b\}$ este gu. ∅

⑧ $x * y$ și $u * u^{-1}$

SOL

$\emptyset$

$\emptyset$

$x = u$

$y = u^{-1}$, $x = u$

REZOLVĂ

$x * y = u * u^{-1}$

$x = u$, $y = u^{-1}$

$y = u^{-1}$

$\emptyset$

OP

DESC.

REZOLVĂ

REZOLVĂ

STOP

$V = \{ y / u^{-1}, x / u \}$ sau

---

⑦ $(x * y) * z$, $u * u^{-1}$

SOL

$\emptyset$

$\emptyset$

~~$(x * y) = u$~~

$* \; u = (x * y)$

$z = (x * y)^{-1}$, $u = (x * y)$

REZOLVĂ

$(x * y) * z = u * u^{-1}$

$(x * y) = u$, $z = u^{-1}$

$z = (x * y)^{-1}$

$\emptyset$

OP

DESC.

~~DE~~ REZOLVĂ

~~REZ~~

STOP

$V = \{ z / (x * y)^{-1}, u / (x * y) \}$ sau

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(1S)

$f(x,y)$, $f(h(x),x)$, $f(x,b)$

SOL

REZOLVA

OP

DESC

$\emptyset$

$f(x,y) = f(h(x)x))$

$f(h(x),x) = f(x,b)$

ESEC

$x = h(x)$, $y = x$

$\emptyset$

$\boxed{h(x) = x}$, $x = b$

(2S) $f(x, f(x, g(y)))$, $f(u,z)$, $f(g(y), y)$

SOL

OP.

DESC

REZ

$f(x, f(x, g(y))) = f(u,z)$

$f(u,z) = f(g(y), y)$

$\emptyset$

REZ

$x = u$, $f(x, g(y)) = z$

$u = g(y)$, $z = f(g(y), y)$

$f(u, g(y)) = z$

$u = g(y)$, $z = y$

REZ

$f(u, g(y)) = y$

$u = g(y)$

REZ

ESEC

$z = y$, $x = u$

5) $f(f(x,y),x)$, $f(g(y),z)$, $f(u,h(z))$

SOL

REZ

$f(f(x,y),x) = f(g(y),z)$

$f(g(y),z) = f(u,h(z))$

$\emptyset$

$f(x,y) = g(y)$, $x = z$

$g(y) = u$, $z = h(z)$

SAU

DESC.

ESEC

↳ fcții diferite
variabile sunt
diferite

4S) $f(f(x,y),x)$, $f(v,u)$, $f(u,h(z))$

SOL

REZ

$f(f(x,y),x) = f(u,v)$

$f(v,u) = f(u,h(z))$

$\emptyset$

OPERAȚII
DESCOMPUNE

$\emptyset$

$f(x,y) = v$, $x = u$

$v = u$, $u = h(z)$

REZOLVĂ

$x = u$

$f(u,y) = v$

$v = u$, $u = h(z)$

REZOLVĂ

$v = f(u,y)$, $x = u$

~~$u = f(u,y)$~~

$f(u,y) = u$, $u = h(z)$

EȘEC.

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(5S) $f(f(x,y),x),\ f(v,u),\ f(u,z)$

SOL	REZ	OP
∅	$f(f(x,y),x) = f(v,u)$ $f(v,u) = f(u,z)$	DESC
∅	$f(x,y) = v,\ x = u$ $v = u,\ u = z$	REZOLVA
$x = u$	$f(u,y) = v$ $v = u,\ u = z$	REZOLVA
$v = u,\ x = u$	$f(u,y) = u$ $u = z$	EȘEC

⇒ B) unificator

(6S) $f(f(g(x),h(y)),h(z)),\ f(f(u,h(h(x))),h(y)),\ f(v,w)$

SOL	REZ	OP
∅	$f(f(g(x),h(y)),h(z)) = f(f(u,h(h(x))),h(y))$ $f(f(u,h(h(x))),h(y)) = f(u,w)$	DESC
∅	$f(g(x),h(y)) = f(u,h(h(x))),\ h(z) = h(y)$ $f(u,h(h(x))) = v,\ h(y) = w$	DESC
∅	$g(x) = u,\ h(y) = h(h(x)),\ h(z) = h(y)$ $f(u,h(h(x))) = v,\ h(y) = w$	REZOLVA
$u = g(x)$	$h(y) = h(h(x)),\ h(z) = h(y)$ $f(g(x),h(h(x))) = v,\ h(y) = w$	REZ

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TEMA

YS $p(x,x,z)$, $p(f(a,a),y,y)$, $p(f(x,a),b,z)$

x, y, z, u, v, w - variabile
a, b, c - constante

SOL	REZ	OP
$\emptyset$	$p(x,x,z) = p(f(a,a),y,y)$ $p(f(a,a),y,y) = p(f(x,a),b,z)$	DESC
$\emptyset$	$x = f(a,a)$, $x = y$, $z = y$ $f(a,a) = f(x,a)$, $y = b$, $y = z$	REZOLVĂ
$x = y$	$y = f(a,a)$, $z = y$ $f(a,a) = f(y,a)$, $y = b$, $y = z$.	REZOLVĂ
$z = y$, $x = y$	$y = f(a,a)$ $f(a,a) = f(y,a) \Rightarrow y = b$, $y = y$	SCOATE
$z = y$, $x = y$	$y = f(a,a)$ $f(a,a) = f(y,a)$, $y = b$	REZOLVĂ
$y = b$, $z = b$, $x = b$	$b = f(a,a)$ $f(a,a) = f(b,a)$	

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(8S) $p(x,x,z), p(f(a,a),y,y), R(x,b,z)$

SOL	REZ	
$\emptyset$	$p(x,x,z) = p(f(a,a),y,y)$ $p(f(a,a),y,y) = p(x,b,z)$	DESC.
$\emptyset$	$x = f(a,a), x = y, z = y$ $f(a,a) = x, y = b, y = z$	REZOL.
$z = y$	$x = f(a,a), x = y$ $f(a,a) = x, y = b, y = y$	SCOATE
$z = y$	$x = f(a,a), x = y$ $f(a,a) = x, y = b$	REZOLVĂ
$x = y$, $z = y$	$y = f(a,a)$ $f(a,a) = y, y = b$	REZOLVĂ
$y = b$, $x = b$, $z = b$	$b = f(a,a)$ $f(a,a) = b$	ESEC.

$\Rightarrow$ ~~X~~ unificator

(9S) $p(x,x,z), p(f(a,a),y,y), R(x,f(a,a),z)$

SOL	REZ	OP
$\emptyset$	$p(x,x,z) = p(f(a,a),y,y)$ $p(f(a,a),y,y) = p(x,f(a,a),z)$	DESC.
$\emptyset$	$x = f(a,a), x = y, z = y$ $f(a,a) = x, y = f(a,a), y = z$	~~DESC.~~ REZOLVĂ
~~$z = y$~~	$x = f(a,a), x = y$ $f(a,a) = x, y = f(a,a), y = y$	SCOATE
$z = y$	$x = f(a,a), x = y$ $f(a,a) = x, y = f(a,a)$	REZOLVĂ

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SOL	REZ	OP
$=y, z=y$	$y=f(a,a)$ $f(a,a)=y, y=f(a,a)$	REZOLVĂ
$y=f(a,a)$, $x=f(a,a)$, $z=f(a,a)$	$f(a,a)=f(a,a)$, $f(a,a)=f(a,a)$	SCOATE
$y=f(a,a)$, $x=f(a,a)$, $z=f(a,a)$	$\emptyset$	STOP.

$\vartheta = \{y \mid f(a,a), x \mid f(a,a), z \mid f(a,a)\}$ ~~out~~ sau.

(10S) $p(f(x,a), g(y), z)$, $p(f(a,a), z, u)$, $p(v, u, z)$

SOL	REZ	OP
$\emptyset$	$p(f(x,a), g(y), z) = p(f(a,a), z, u)$ $p(f(a,a), z, u) = p(v, u, z)$	DESC
$\emptyset$	$f(x,a) = f(a,a)$, $g(y) = z$, $z = u$ $f(a,a) = v$, $z = u$, $u = z$	DESC.
$\emptyset$	$x = a$, $a = a$, $g(y) = z$, $z = u$ $f(a,a) = v$, $z = u$, $u = z$	SCOATE
$\emptyset$	$x = a$, $g(y) = z$, $z = u$ $f(a,a) = v$, $z = u$, $u = z$	REZOLVĂ
$y = a$	$g(y) = z$, $z = u$ $f(a,a) = v$, $z = u$, $u = z$	REZOLVĂ

→

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$z = u$, $x = a$	$g(y) = u$, $f(a,a) = v$, $u = u$, $u = u$	SCOATE
$z = u$, $x = a$	$g(y) = u$ $f(a,a) = v$	REZOLVĂ
$u = g(y)$, $z = g(y)$, $x = a$	$f(a,a) = v$	REZOLVĂ
$v = f(a,a)$, $u = g(y)$, $z = g(y)$, $x = a$	$\emptyset$	STOP

$\Rightarrow \gamma = \{v \mid f(a,a),\ u \mid g(y),\ z \mid g(y),\ x \mid a\}$

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Examen

$\{h(x, y, f(g(x), g(y))) = h(y, x, f(y, x))\}$

SOL	REZ	OP
$\emptyset$	$h(x, y, f(g(x), g(y))) = h(y, x, f(y, x))$	DESC
$\emptyset$	$x = y, y = x, f(g(x), g(y)) = f(y, x)$	DESC
$\emptyset$	$x = y, y = x, g(x) = y, g(y) = x$	REZ.
$\{y = y\}$	$y = y, g(y) = y, g(y) = y$	~~ESEC~~ SCOATE
$\{g(y) = y\}$	$g(y) = y, g(y) = y$	ESEC.

TEMA: toate ex. unificare.

I. SLD

~~R A R~~

$(P \wedge Q) \rightarrow R$

"?-" → ce vrem să arăt.

$r :- p, q. = (p \wedge q) \rightarrow r$

$r :- p, q. = \neg p \vee \neg q \vee r$

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SOL	REZ	OP
$w = h(y)$, $u = g(x)$	$h(y) = h(h(x))$, $h(z) = h(y)$ $f(g(x), h(h(x))) = v$	REZ.
$v = f(g(x), h(h(x)))$, $w = h(y)$, $u = g(x)$	$h(y) = h(h(x))$, $h(z) = h(y)$	REZ DESC.
~~$h(z) =$~~ ~~$h(y) = h(h(x))$~~, $v = f(g(x), h(h(x)))$, $w = h(y)$, $u = g(x)$	~~$h(h(x))$~~ ~~$h(z)$~~ $y = h(x)$, $z = y$	REZ.
$z = y$, $v = f(g(x), h(h(x)))$, $w = h(y)$, $u = g(x)$	$y = h(y)$	REZ.
$y = h(x)$, $z = h(x)$, $v = f(g(x), h(h(x)))$, $w = h(h(x))$, $u = g(x)$	$\emptyset$	STOP

$V = \{y \mid h(x),\ z \mid h(x),\ v \mid f(g(x)), h(h(x)),\ w \mid h(h(x)),\ u \mid g(x)\}$ gen.

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(15) r :- p, q.
s :- p, q.
v :- t, u.
w :- v, s.
t.
q.
u.
p.

(1) $\neg p \vee \neg q \vee r$
(2) $\neg p \vee \neg q \vee s$
(3) $\neg t \vee \neg u \vee v$
(4) $\neg v \vee \neg s \vee w$
(5) t
(6) q
(7) u
(8) p

? - w.

$G_0 = \neg w$

$G_1 = \neg v \vee \neg s$ (R4)

$G_2 = \neg v \vee \neg p \vee \neg q$ (R2)

$G_3 = \neg v \vee \neg p$ (R6)

$G_4 = \neg v$ (R8)

$G_5 = \neg t \vee \neg u$ (R3)

$G_6 = \neg u$ (R5)

$G_7 = \square$ (R7)

L1

$\neg v \vee \neg s$ +
$\neg p \vee \neg q \vee s$
$\neg v \vee \neg p \vee \neg q$

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(23) $q(x,y) :- q(y,x), q(y, f(f(y)))$.

$q(a, f(f(x)))$.

$?- q(f(z), a)$.

(1) ~~$(q(y,x) \wedge q($~~ $\equiv (1) \neg(q(y,x)) \vee \neg q(y, f(f(y)))$ $\Rightarrow \vee q(x,y)$

(2) $q(a, f(f(x)))$

$G_0 = \neg q(f(f(z)), a)$ R1 cu $\theta(y) = f(z)$, $\theta(y) = a$

~~$G_1 = \neg q(a, f(z)) \vee \neg q(a, f(f(f(z))))$~~

$G_1 = \neg q(a, f(z)) \vee \neg q(a, f(f(a)))$

$G_2 = \neg q(a, f(z))$ R2 cu $(\theta(x) = a)$

$G_3 = \square$ R2 cu $(\theta(z) = f(x))$

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(35) p(X) :- q(x, f(y)), r(a).
p(x) :- r(a).
q(x, y) :- p(y).
r(x) :- q(x, y).
r(f(b)).

?- p(x), q(y, z).

(1) ¬(q(x, f(y)) ∨ ¬r(a) ∨ p(x)
(2) ¬r(x) ∨ p(x)
(3) ¬p(y) ∨ q(x, y)
(4) ¬q(x, y) ∨ r(x)
(5) r(f(b))

Atenție! Prioritizez variabilele din ~~ipoteză~~ în cazul în care fac schimbare NUMAI altre variabile

$G_0 = \neg(p(x)) \vee \neg q(y, z)$
$G_1 = \neg q(y, z) \vee \neg r(x)$
$G_2 = \neg q(y, z)$
$G_3 = \neg p(z)$
~~$G_4 = \neg r(z)$~~
$G_3 = \neg p(z)$
$G_4 = \neg r(x)$
$G_5 = \square$

(R2) =>
($\theta(x) = f(b)$ R_5)
($\theta(x) = y$ → doar variabile
~~$\theta(y) = z$~~)
cu R_3
(R_2 cu $\theta(z) = z$)

(R_5 cu $\theta(x) = f(b)$)

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Examen

(1) shuffle(lit(x), lit(y))

(2) ¬(shuffle(x, u)) ∨ ¬(shuffle(y, w)) ∨ ¬(shuffle(z, t)) ∨ shuffle(arb(x, y, z), arb(T, u, w))

?- shuffle(x, arb(arb(lit(t), lit(c), lit(u)), arb(lit(i), lit(a), lit(t)), arb(lit(e), lit(t), lit(r))))

$G_0 = \neg(shuffle(x_1, arb(arb(lit(t), lit(c), lit(u)), arb(lit(i), lit(a), lit(t)), arb(lit(e), lit(t), lit(r)))))$.

$G_1 = \neg s(x_1) \vee \neg s(y_1) \vee \neg s(z_1) \vee$

a(l(i), l(a), l(t)) a(l(e), l(t), l(r)) a(l(t), l(c), l(u))

* R2 con $\theta(X_1) = arb(x, y, z)$

$\theta(T) = arb(lit(t), lit(c), lit(u))$

$\theta(u) = arb(lit(i), lit(a), lit(t))$

$\theta(w) = arb(lit(e), lit(t), lit(r))$

~~$G_2 = \neg s(y_1) \vee \neg s(z_1) \vee$~~

~~R1 con $\theta(x) = l(x)$~~

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NIFICARE - ex. Macovei

9) $x * y \;,\; x * (y * (u * v)^{-1})$

SOL	REZOLVARE	OP
∅	$x * y = x * (y * (u * v)^{-1})$	DESC
∅	$x = x,\ y = (y * (u * v)^{-1})$	SCOATE
∅	$y = (y) * (u * v)^{-1}$	EȘEC.

$\Rightarrow \nexists$ unificator

(10) $x * y \;,\; y * (u * v)^{-1}$

SOL	~~REZ~~	OP
∅	$x * y = y * (u * v)^{-1}$	DESC.
∅	$x = y,\ y = (u * v)^{-1}$	REZOLVĂ.
$x = y$	$y = (u * v)^{-1}$	REZOLVĂ

$y = (u * v)^{-1}$
$x = (u * v)^{-1}$ ∅ STOP

$\Rightarrow V = \{y \mid (u * v)^{-1},\ x \mid (u * v)^{-1}\}$ este CGU.

(11) $f(g(x), x) \;,\; f(y, y)$

SOL	REZ	OP
∅	$f(g(x), x) = f(y, y)$	DESC
∅	$g(x) = y,\ x = y$	REZ.
$x = y$	$g(y) = y$	EȘEC.

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(14) $f(x, f(b,x)), f(f(y,a), f(b, f(z,z)))$

SOL	REZ	OP
$\emptyset$	$f(x, f(b,x)) = f(f(y,a), f(b, f(z,z))$	DE
$\emptyset$	$x = f(y,a), f(b,x) = f(b, f(z,z))$	DESC
$\emptyset$	$x = f(y,a), b = b, x = f(z,z)$	SCOATE
$\emptyset$	$x = f(y,a), x = f(z,z)$	REZOLVĂ
$x = f(z,z)$	$f(z,z) = f(y,a)$	DESC
$x = f(z,z)$	$z = y, z = a$	REZOLVĂ
$z = y,$ $x = f(y,y)$	$y = a$	REZOLVĂ
$y = a,$ $z = a,$ $x = f(a,a)$	$\emptyset$	STOP

$\Rightarrow \sigma = \{y/a, z/a, x/f(a,a)\}$

(15) $p(x,b,x), p(y,y,c)$

SOL	REZ	OP
$\emptyset$	$p(x,b,x) = p(y,y,c)$	DESC
$\emptyset$	$x = y, b = y, x = c$	REZOLVĂ
$x = y$	$b = y, y = c$	REZOLVĂ
$y = b,$ $x = b$	$b = c$	EȘEC.

$\Rightarrow$ ✗ unificator.

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$f(x,y)$, $f(h(x),x)$, $f(x,b)$

SOL — REZ — OP

$\emptyset$ — $f(x,y) = f(h(x),x)$ — DESC

$f(h(x),x) = f(x,b)$

$\emptyset$ — $x = h(x)$, $y = x$ — ESEC.

$h(x) = x$, $x = b$

$\Rightarrow$ ~~X~~ unificator

(4) $f(x, f(x, g(y)))$, $f(u,z)$, $f(g(y), y)$

SOL — REZ — OP

$\emptyset$ — $f(x, f(x,g(y))) = f(u,z)$ — DESC.

$f(u,z) = f(g(y), y)$

$\emptyset$ — $x = u$, $f(x, g(y)) = z$ — REZOLVĂ

$u = g(y)$, $z = y$

$x = u$ — $f(u, g(y)) = z$ — REZOLVĂ

$u = g(y)$ $z = y$

$z = y$, $x = u$ — $f(u, g(y)) = y$ — ESEC

$u = g(y)$

$\Rightarrow$ ~~X~~ unificator.

(8) $f(f(x,y), x)$, $f(g(y), z)$, $f(u, h(z))$

SOL — REZ — OP

$\emptyset$ — $f(f(x,y), x) = f(g(y), z)$ — DESC

$f(g(y), z) = f(u, h(z))$

$\emptyset$ — $f(x,y) = g(y)$, $x = z$ — ESEC

$g(y) = u$, $z = h(z)$ — $(f \neq g)$

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(19) $f(f(x,y),x),\ f(v,u),\ f(u,h(z))$

SOL	REZOLVARE	OP
∅	$f(f(x,y),x) = f(v,u)$ $f(v,u) = f(u,h(z))$	DESC
∅	$f(x,y) = v,\ x = u$ $v = u,\ u = h(z)$	REZOLVARE
$v = u$	$f(x,y) = u,\ x = u$ $u = h(z)$	REZOLVARE
$x = u,$ $v = u$	$f(u,y) = u$ $u = h(z)$	ESEC

$\Rightarrow$ ~~x~~ unificator

model examen 2022-2023

$f(x, g(x), h(a, g(y)))\ ,\ f(a, g(x), z),$
$f(y, y, h(a, z))$

SOL	REZ	OP
∅	$f(x, g(x), h(a, g(y))) = f(a, g(x), z)$ $f(a, g(x), z) = f(y, y, h(a, z))$	DESC
∅	$x = a,\ g(x) = g(x),\ h(a, g(y)) = z$ $a = y,\ g(x) = y,\ z = h(a, z)$	SCOATE
∅	$x = a,\ h(a, g(y)) = z$ $a = y,\ g(x) = y,\ z = h(a, z)$	ESEC.

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Model examen 2022-2023

SLD - rezolvare

(1) $m(a,b)$.

(2) $f(a,b)$.

(3) $p(a)$.

(4) $p(X) :- f(Y,X), p(Y)$.

$?- p(X), m(Y,X)$.

---

(1) $m(a,b)$

(2) $f(a,b)$

(3) $p(a)$

(4) $\neg f(Y,X) \vee \neg p(Y) \vee p(X)$

$G_0 = \neg p(X_1) \vee \neg m(Y_1, X_1)$

$G_1 = \neg m(Y_1, X_1) \vee \neg f(Y_1, X_1) \vee \neg p(Y_1)$ (R_4 cu $\theta(X_1) = X$, $\theta(Y_1) = Y$)

~~$G_2 = \neg f(Y_1, X_1) \vee \neg p(Y_1)$~~ ($R_1$ cu $\theta(Y_1) = a$, $\theta(X_1) = b$)

$G_2 = \neg f(Y, X) \vee \neg p(Y)$

~~$G_3 =$~~

$G_0 = \neg p(X_1) \vee \neg m(Y_1, X_1)$

$G_1 = \neg m(Y, X) \vee \neg f(Y, X) \vee \neg p(Y)$ (R_4 cu $\theta(X_1) = X$, $\theta(Y_1) = Y$)

$G_2 = \neg f(a,b) \vee \neg p(a)$ (R_1 cu $\theta(Y) = a$, $\theta(X) = b$)

$G_3 = \neg f(a,b)$ (R_3)

$G_4 = \square$ (R_2)

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(11) SLD.

(1) $p(X,Y) :- q(X,Z), r(Z,Y).$
(2) $p(X,X) :- s(X).$
(3) $q(X,b).$
(4) $q(b,a).$
(5) $q(X,a) :- r(a,X).$
(6) $r(b,a).$
(7) $s(X) :- t(X,a).$
(8) $s(X) :- t(X,b).$
(9) $s(X) :- t(X,X)$
(10) $t(a,b)$
(11) $t(b,a)$

?- $p(X,X)$

---

(1) $\neg q(X,Z) \vee \neg r(Z,Y) \vee p(X,Y).$
(2) $\neg s(X) \vee p(X,X).$
(3) $q(X,b).$
(4) $q(b,a)$
(5) $\neg r(a,X) \vee q(X,a)$
(6) $r(b,a)$
(7) $\neg t(X,a) \vee s(X)$
(8) $\neg t(X,b) \vee s(X)$
(9) $\neg t(X,X) \vee s(X)$
(10) $t(a,b)$
(11) $t(b,a)$

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$G_0 = \neg p(X_1, X_1)$

$G_1 = \neg s(X)$ $\quad (R_2 \text{ cu } \theta(X_1) = X)$

$G_2 = \neg t(X, a)$ $\quad (R_4)$ ~~cu θ~~

$G_3 = \square$ $\quad (R_{11} \text{ cu } \theta(X) = b)$

Seminar

(1) $m(e, c)$

(2) $m(d, b)$

(3) $f(a, b)$

(4) $f(a, c)$

(5) $p(a)$

(6) $p(d)$

(7) $p(X) :- f(Y, X), p(Y) \Rightarrow \neg f(Y, X) \vee \neg p(Y) \vee p(X)$

$? - p(X), m(Y, X), p(Y)$

~~$G_0 = \neg p(X) \vee \neg m(Y_1, X_1) \vee \neg p(Y_1)$~~

~~$G_1 = \neg m(Y_1, a) \vee \neg p(Y_1)$~~ $\quad (R_5 \text{ cu } \theta(X_1) = a)$

~~$G_2 = \neg m(d, a)$~~ $\quad (R_6 \text{ cu } \theta(Y_1) = d)$

$G_0 = \neg p(X_1) \vee \neg m(Y_1, X_1) \vee \neg p(Y_1)$

$G_1 = \neg m(Y_1, X) \vee \neg p(Y_2) \vee \neg f(Y, X) \vee \neg p(Y)$ $\quad (R_7 \text{ cu } \theta(X_1) = X, \theta(Y_1) = Y)$

$G_2 = \neg m(d, X) \vee \neg p(d) \vee \neg f(d, X)$ $\quad (R_6 \text{ cu } \theta(Y) = d)$

$G_3 = \neg m(d, X) \vee \neg f(d, X)$ $\quad (R_6)$ ~~θ~~

$G_4 = \neg f(d, b)$ $\quad (R_2 \text{ cu } \theta(X) = b)$

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$G_0 = \neg P(X_1) \vee \neg m(Y_1, X_1) \vee \neg P(Y_1)$

$G_1 = \neg P(X_1) \vee \neg m(d, X_1)$

$G_2 = \neg m(d, X) \vee \neg$

$(R_6 \; m \; \theta(Y_1) = d)$

- $G_0 = \neg P(X_1) \vee \neg m(Y_1, X_1) \vee \neg P(Y_1)$

  $(R_2 \; m \; \theta(Y_1) = a$
  $\theta(X_1) = b$

  $G_1 = \neg P(b) \vee \neg P(d)$ (R_6)

  $(R_7 \; m \; \theta(X) = b)$

  $G_2 = \neg P(b)$

  $G_3 = \neg f(y, b) \vee \neg P(Y)$

  $(R_5 \; m \; \theta(Y) = a)$

  $G_4 = \neg f(a, b)$

  $G_5 = \square$ (R_3)

- $G_0 = \neg P(X_1) \vee \neg m(Y_1, X_1) \vee \neg P(Y_1)$

  $(R_6 \; m \; \theta(Y_1) = d)$

  $G_1 = \neg P(X_1) \vee \neg m(d, X_1)$

  $(R_2 \; m \; \theta(X_1) = b)$

  $G_2 = \neg P(b)$

  $G_3 = \neg P(b) \vee \neg f(Y, b) \vee \neg P(Y)$ $(R_7 \; m \; \theta(X) = b)$

  $(R_5 \; m \; \theta(Y) = a$

  $G_4 = \neg f(a, b)$

  $G_5 = \square$ (R_3)

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Seminar 6 FLP

λ - calcul

$f(x) = t$
$x \to t$

Sintaxa $t ::= x \mid t\ t \mid \lambda x.t$

variabilă aplicare abstractizare

$\lambda x.\ x^2+1 \Rightarrow f(x) = x^2+1$

Fie Var_Λ o mulțime numărabilă.
În general, notăm variabilele cu $\{x, y, z, \dots\}$

Atunci, ΛT este:
- $Var_\Lambda \subseteq \Lambda T$
- dacă $t_1, t_2 \in \Lambda T \Rightarrow (t_1 t_2) \in \Lambda T$
- dacă $x \in Var_\Lambda$ și $t \in \Lambda T \Rightarrow (\lambda x.t) \in \Lambda T$

- Aplicarea e asociativă la stânga
  $t_1 t_2 t_3 = (t_1 t_2) t_3$
- Corpul abstractizării e extins la dreapta
  $\lambda x.\ t_1 t_2 = \lambda x.\ (t_1 t_2)$
- $\lambda xyz.t = \lambda x.\ \lambda y.\ \lambda z.\ t$

$\lambda x.t$; aparițiile lui x în t sunt legate dacă $y \in t$, $y \neq x$, y este var. liberă

$FV : \Lambda T \to \mathcal{P}(Var_\Lambda)$
$FV(x) = \{x\}$
$FV(t_1 t_2) = FV(t_1) \cup FV(t_2)$
$FV(\lambda x.t) = FV(t) \setminus \{x\}$

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① Calculați

a) $FV(\lambda x.xy) = FV(xy) \setminus \{x\}$
$= FV(x) \cup FV(y) \setminus \{x\}$
$= \{x\} \cup \{y\} \setminus \{x\}$
$= \{y\}$

b) $FV(x\lambda x.\lambda y) = FV(x) \cup FV(\lambda x.xy)$
$= \{x\} \cup \{y\} = \{x, y\}$

c) $FV(x(\lambda xy.xyz)(\lambda v.yv)) =$
$= \{x, z, y\}$

d) $FV(\lambda t.((\lambda xyz.yzx)t)) = \emptyset$

Obs! În cazul în care pentru un $t \in \Lambda T$ avem $FV(t) = \emptyset$ spunem că t este termen închis.

Relația $=_\alpha$ (α-echivalență, α-conversie)

$=_\alpha \subseteq \Lambda T^2$ (relație de echivalență, reflexivă + compatibilitate)

(refl) $t =_\alpha t$

(sim) $t_1 =_\alpha t_2 \Rightarrow t_2 =_\alpha t_1$

(trans) $t_1 =_\alpha t_2$ și $t_2 =_\alpha t_3 \Rightarrow t_1 =_\alpha t_3$

(redenumire) $\lambda x.t =_\alpha \lambda y.[y/x]t$ unde $y \neq x$ și $x \notin FV(y)$

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(compatibilizate) $t_1 =_\alpha t_2 \to$ $t\,t_1 =_\alpha t\,t_2$
$t_1\,t =_\alpha t_2\,t$
$\lambda x.t_1 =_\alpha \lambda x.t_2$

$f(x) = x+1$
$f(y) = y+1$ $\quad \lambda x.x+1 =_\alpha \lambda y.y+1$

Substitutia

(Var) $[u/x]x = u$
(Var) $[u/x]y = y$
~~(Var)~~ (Ap) $[u/x](t_1 t_2) = ([u/x]t_1)([u/x]t_2)$
(Abstr) $[u/x](\lambda x.t) = \lambda x.t$
(Abstr) $[u/x](\lambda y.t) = \lambda y.[u/x]t$
$y \neq x$, $y \in FV(u)$
(Abstr) $[u/x](\lambda y.t) = \lambda z.[u/x]([z/y]t)$
$y \neq x$, $y \in FV(u)$, z var. nouă

$M, \rho \subseteq M^2$, $M/\rho := \{x' \mid x \in M\}$
$\equiv_n \subseteq \mathbb{Z}^2$, $\mathbb{Z}/\equiv_n := \mathbb{Z}_n$
$\mathbb{Z}_n = \{\hat{0}, \hat{1}, \dots, \hat{n-1}\}$
$\mathbb{Z}_5 : \hat{0} \equiv_5 \hat{5}$

$\Lambda T := (\Lambda T/=_\alpha)$
$\lambda x.xy =_\alpha \lambda x.\lambda y$

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② a) $[y|x]\lambda z.x = \lambda z.y$

b) $[y|x]\lambda y.x =_\alpha [y|x]\lambda z.x =_\alpha \lambda z.y$

c) $[\lambda z.z|x](\lambda y.yx) =_\alpha \lambda x.yx$

d) $[\lambda z.z|x](\lambda y.yx) = \lambda y.y(\lambda z.z)$

$\to_\beta$, $\xrightarrow{*}_\beta$ (star Kleene)

$\to_\beta$, $\xrightarrow{*}_\beta \subseteq (\Lambda T/=_\alpha)^2$ calculul functional

- aplicare $(\lambda x.t)v = [v|x]t$
- compatibilitate: dacă $t_1 \to_\beta t_2$ atunci $\begin{cases} tt_1 \to_\beta tt_2 \\ t_1 t \to_\beta t_2 t \\ \lambda x.t_1 \to_\beta \lambda x.t_2 \end{cases}$

$(\lambda x.x+1)2 \to_\beta [2|x]x+1 =_\alpha 2+1 =_\alpha 3$

Def $\xrightarrow{*}_\beta$ $t_1 \xrightarrow{*}_\beta t_2$ dacă $\exists v_0, \dots v_n \in \Lambda T$ cu $n \geq 0$ a.î. $t_1 =_\alpha v_0 \to_\beta v_1 \to_\beta \dots \to_\beta v_n =_\alpha t_2$

③ a) $\lambda x.(\lambda y.yx)z)v \to_\beta [v|x](\lambda y.yx)z$

$=_\alpha (\lambda y.yv)z \to_\beta [z|y]yv =_\alpha zv$

b) $\lambda x.((\lambda y.yx)z))v \to_\beta (\lambda x.[z|y]yx)v =_\alpha$

$(\lambda x.zx)v \to_\beta [v|x]zx =_\alpha zv$

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Sistemul de rescriere prin $\to_\beta$ e confluent.

Teorema Church - Rosser

Dacă $t \in \Lambda T$ și $t \to_\beta^* t_1$, $t \to^* t_2 \Rightarrow \exists\, u \in \Lambda T$ a.î. $t_1 \to_\beta^* u$ și $t_2 \to_\beta^* u$

Obs! Sistemul de rescriere $\to_\beta$ nu este Noetherian.

Noetherian → rescriere finită

Dacă u nu se mai poate rescrie, atunci il numim β-forma normală a lui t_i.

Reprezentarea aritmetică prin numerale Church

Axiomatica lui Peano
$(\mathbb{N}, succ, 0)$
$0 \in \mathbb{N}$
$1 := succ\ 0$
$2 := succ\ 1 \equiv succ\ (succ\ 0)$
$n := succ^n\ 0$

$0 := \lambda f x.\ x$
$1 := \lambda f x.\ f\ x$
$2 := \lambda f x.\ f\ (f\ x)$
$\vdots$
$n := \lambda f x.\ f^n\ x$

$succ \equiv \lambda n f x.\ f\ (n\ f\ x)$

$succ\ 2 =_\alpha (\lambda n f x.\ f\ (n\ f\ x))\ (\lambda f x.\ f\ (f\ x))$
$\to_\beta (\lambda f x.\ f\ (f\ x))\ n\ f\ (\lambda f x.\ f\ (n\ f\ x))$
$=_\alpha \lambda f x.\ f\ ((\lambda f x.\ f\ (f\ x))\ f\ x) \to_\beta^*$

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$\xrightarrow{*}_\beta \lambda fx.\, f(f(fx)) \equiv 3$

$add \equiv \lambda mnfx.\, mf.(nfx)$

$add\ 0\ 2 \equiv (\lambda mnfx.\, mf(nfx))\ (\lambda fx.\, x)$

$(\lambda fx.\, f(fx)) \xrightarrow{*}_\beta [m \mid \lambda fx.\, x]\,[n \mid \lambda f.\, f(fx)]$

$\lambda fx.\, mf(nfx) =_\alpha \lambda fx.\, (\lambda fx.\, x)f\,((\lambda fx.\, f(fx))f$

$\xrightarrow{*}_\beta \lambda fx.\, (\lambda fx.\, x)f(f(fx)) \xrightarrow{*}_\beta \lambda fx.\, f(fx) \equiv 2$

Tema

① $add\ 1\ 2 = 3$

② $add\ 3\ 5 = 8$

③ $succ\ 3 = 4$

④ ~~succ~~ $add\ 1 =_\alpha succ$

$add\ 1\ 2 = \lambda mnfx.\, mf(nfx)\,(\lambda fx.\, fx)\,(\lambda fx.\, f(fx))$

$\xrightarrow{}_\beta [m \mid \lambda fx.\, fx]\,[n \mid \lambda fx.\, f(fx)]$

$\lambda fx.\, mf(nfx) =_\alpha \lambda fx.\, (\lambda fx.\, fx)f(\lambda fx.\, f(fx))$

~~$=_\alpha \lambda fx.\, f(f(f(fx))) \equiv 3$~~

$=_\alpha \lambda fx.\, f(f(f(x))) \equiv 3$

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Recap - Macovei FLP

Unificare → cazuri eșec : • $x = t \;/\; t = x \leftrightarrow x \in T$ (ex. $f(x) = x$)

• $f(x) = g(y) \leftrightarrow f \neq g$

• $f(x) = f(v,v) \leftrightarrow f$ de arități diferite

① a) $f(x,y)$, $f(h(x),x)$ și $f(x,b)$

Soluție	Mulțimea de rezolvat	Operație
	$f(x,y) = f(h(x),x)$ și $f(h(x),x) = f(x,b)$	Descompune
$\emptyset$	$x = h(x)$, $y = x$ și $h(x) = x$, $x = b$	Eșec

→ F unificator pentru termenii dați

b) $f(f(x,y),x)$, $f(v,u)$ și $f(u,h(z))$

Soluție	Mulțime de rezolvat	Operație
	$f(f(x,y),x) = f(v,u)$ și $f(v,u) = f(u,h(z))$	Desc.
$\emptyset$	$f(x,y) = v$, $x = u$ și $v = u$, $u = h(z)$	Rezolvă
$v = f(x,y)$	$x = u$, $f(x,y) = u$, $u = h(z)$	Rezolvă
~~$x = u$~~, $v = f(u,y)$	$f(v,y) = v$, $u = h(z)$	Eșec

→ F unificator pentru termenii dați

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c) $p(x,x,z)$, $p(f(a,a),y,y)$?: ~~$p(f(x,a),f,y,y)$~~ $p(x,b,z)$

Soluție	Mulțime de rezolvat	Soluție
		1
~~$\emptyset$~~	$p(x,x,z)=p(f(a,a),y,y)$; $p(f(a,a),y,y)=p(x,b,z)$	
~~$\emptyset$~~	$x=f(a,a)$, $x=y$, $z=y$; $f(a,a)=x$, $y=b$, $y=z$	R
$x=f(a,a)$	$f(a,a)=y$, $z=y$; $y=b$, $y=z$ ($f(a,a)=f(a,a)$)	S
$x=f(a,a)$	$f(a,a)=y$, $z=y$, $y=b$, $y=z$	R
~~$y=f(a,a)$~~ $z=y$, $x=f(a,a)$	$f(a,a)=y$, $y=y$, $y=b$, $y=y$	S
$z=y$, $x=f(a,a)$	$f(a,a)=y$, $y=b$	R
$y=f(a,a)$, $z=f(a,a)$, $x=f(a,a)$	$f(a,a)=b$	Eșec

⇒ ∄ unificator pentru termenii dați

! Dacă aveam în loc de b, o var. u ⇒

~~$y=b$, $z=b$, $x=$~~		
$y=f(a,a)$, $z=f(a,a)$, $x=f(a,a)$	$f(a,a)=u$	R
$u=f(a,a)$, $y=f(a,a)$, $z=f(a,a)$, $x=f(a,a)$	$\emptyset$	Stop

∃ unificator ⇒ $\sigma = \{f(a,a)/u, f(a,a)/z, f(a,a)/x, f(a,a)/y\}$

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Deducție naturală

② a) $p \to q, r \to s \vdash (p \vee r) \to (q \vee s)$

1) $p \to q$ ipoteză

2) $r \to s$ ipoteză

3) $p \vee r$ asumpție

4) p asumpție

5) q $\to e\ (1, 4)$

6) $q \vee s$ $\vee i_1\ (5)$

7) r asumpție

8) s $\to e\ (2, 7)$

9) $q \vee s$ $\vee i_2\ (8)$

10) $q \vee s$ $\vee e\ (3, 4-6, 7-9)$

11) $(p \vee r) \to (q \vee s)$ $\to i\ (3-10)$

b) $\neg q, p \vee q, s \to \neg p, s \vdash r$

1) $\neg q$ ipoteză

2) $p \vee q$ ipoteză

3) $s \to \neg p$ ipoteză

4) s ipoteză

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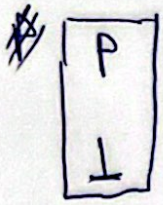
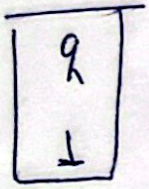
5) $\neg p$ $\rightarrow e\ (3,4)$

6)  asumptie

7) $\neg e\ (5,6)$

8)  asumptie

9) $\neg e\ (1,8)$

10) $\perp$ $\vee e\ (2,6\text{-}7,8\text{-}9)$

11) $\neg r$ $\perp e(10)$

Puncte fixe și Teorema Knaster – Tarski

③
```
a :- b, c, f.
b.
d.
e.
c :- d, b, e.
```

$\Rightarrow S = \{ b \wedge c \wedge f \rightarrow a,\ b,\ d,\ e,\ d \wedge b \wedge e \rightarrow c \}$

$f_S(\emptyset) = \{b, d, e\} \cup \emptyset = \{b, d, e\}$

$f_S(\{b,d,e\}) = \{b,d,e\} \cup \{b,d,e\} \cup \{c\} = \{c,b,d,e\}$

$f_S(\{c,b,d,e\}) = \{c,b,d,e\} \cup \{b,d,e\} \cup \{c\} = \{c,b,d,e\}$

$\Rightarrow lfp_S \rightarrow \{c,b,d,e\} = lfp_S$

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Rezolutie SLD

④ $p(x) :- q(x, f(y)), r(a)$.

$p(x) :- r(x)$.

$q(x,y) :- p(y)$

$r(x) :- q(x,y)$.

$r(f(b))$.

$? - p(x), q(y,z)$

1) $\neg q(x, f(y)) \vee \neg r(a) \vee p(x)$

2) $\neg r(x) \vee p(x)$

3) $\neg p(y) \vee q(x,y)$

4) $\neg q(x,y) \vee r(x)$

5 $r(f(b))$

$G_0 = \neg p(x) \vee \neg q(y,z)$

$G_1 = \neg q(y,z) \vee \neg r(x)$ Rez(2) cu $\theta(x) = x$

$G_2 = \neg q(y,z)$ Rez(5) cu $\theta(x) = f(b)$

$G_3 = \neg p(z)$ Rez(3) cu $\theta(x) = y, \theta(y) = z$

$G_4 = \neg r(z)$ Rez(2) cu $\theta(x) = z$

$G_5 = \square$ Rez cu $\theta(z) = f(b)$

⑤

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λ-calcul

!! $\lambda x.t \Rightarrow f(x) = t$

$\lambda xy.t \Rightarrow f(x,y) = t$

$(\lambda xy.t)ab \rightarrow f(x,y) = t$

$x = a$

$y = b$

⑤ a) true := $\lambda tf.t$

false := $\lambda tf.f$

and := $\lambda pq.pqp$

or := $\lambda pq.ppq$

De aratat → and true false $\longrightarrow_\beta$ false

→ $\forall x \in$ {true, false} (or true x $\longrightarrow_\beta$ true)

and true false := $(\lambda pq.pqp)$ true false $\longrightarrow_\beta$

$\longrightarrow_\beta \lambda tf.$ true false true := $(\lambda tf.t)$ false true $\longrightarrow_\beta$ false

b) 0 := $\lambda fx.x$

1 := $\lambda fx.fx$

2 := $\lambda fx.f(fx)$

3 := $\lambda fx.f(f(fx))$

succ := $\lambda nfx.f(nfx)$

add := $\lambda mnfx.mf(nfx)$

succ 2 : $\lambda nfx.f(nfx)\,2 \longrightarrow_\beta \lambda fx.f(2fx) :=$

$\lambda fx.f(\lambda f'x'.f'(f'(x))fx) \Longrightarrow_\alpha \lambda fx.f(f(fx)) := 3$

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a) or true x :=$(\lambda pq.ppq)$ true x $\longrightarrow_B$ true true x

$\lambda tf.t$(true x) $\longrightarrow_B$ true

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